The Wilken–Weiermann Comparison Problem Foundations and a Guide to the Results

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Abstract

Wilken and Weiermann's analysis of Gödel's T raises two different effectiveness questions: a syntactic termination ordering with specified stability and ordinal calibration, and semantic comparison of the ordinal expressions used by a particular assignment. We explain the distinction and give a common foundation for a collection of proofs concerning both questions. We prove an explicit maximal-predecessor algorithm, an exact single-path evaluator for the collapse, its terminal-count decomposition, and the forward arithmetic form of actual canonical assignments with ground free variables. These results connect typed syntax to an arithmetic problem whose positive and negative boundaries are developed in the companion papers. We explain why complete counts, intermediate block counts and finite-state observers have different decision-theoretic roles. The remaining exact-standard- control comparison problem with several independent ground inputs stays open.

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1 What the problem asks

A termination proof and an algorithm for comparing its measures answer different questions. A reduction sequence can be known to terminate because some ordinal decreases, even when the proposed comparison of two expressions quantifies over infinitely many possible substitutions. Conversely, a decidable ordering on syntax can prove termination without deciding the particular universally quantified numerical relation that motivated it.

Wilken and Weiermann obtain an optimal derivation-length classification for the lambda formulation of Gödel's T [1]. Their Introduction anticipates syntactic termination orderings extending recursive path ordering, with Bachmann–Howard order type. Their effectiveness conjecture following Definition 2.8 concerns comparison of their ordinal expressions and vectors under all admissible substitutions. These are two mathematical contracts. The collection keeps them separate throughout.

Three objects are especially easy to confuse. A general assignment in [1] may attach a finite reduction history to an abstraction; the normalization argument constructs appropriate decreasing assignments along reductions. The *canonical assignment* fixes history length zero at every abstraction. Once its binary macro-expansion convention is fixed, one literal input term determines one vector. Finally, a *syntactic ordering* is a relation defined on terms and stable under substitution and contexts. A result about one object is a result about another only after an explicit interpretation theorem.

The main unresolved domain considered here fixes the source grammar, the zero-history canonical assignment with the explicit literal expansion convention of Section 5, and the numerical control $\Phi(t) = F_5(t + 100)$ defined below. All free variables have ground type; bound variables, recursors, output types and internal simple types are unrestricted. Distinct ground names vary independently. This domain is smaller than the unrestricted expression/substitution problem in the original effectiveness conjecture, but it is still not known to be decidable.

Open problem 1.1 (Exact canonical comparison)**.** Given two finite original T terms with only ground free variables, decide whether their canonical vectors satisfy universal weak comparison, or universal Wilken–Weiermann strict comparison, at the fixed control Φ.

An algorithm must halt on every input pair, including pairs for which there is no countervaluation anywhere. It need not have an elementary or practical running time. An enormous closed integer may therefore be computed as part of a mathematical reduction, but a finite numerical experiment with a smaller control is not an execution of that reduction on its genuine parameters.

2 The calculus and the two comparison questions

The ground type is o. Simple types are formed by $\sigma \to \tau$, with arrows associated to the right, and their levels are

$$\mathrm{lv}(o) = 0, \qquad \mathrm{lv}(\sigma \to \tau) = \max\{\mathrm{lv}(\sigma) + 1, \mathrm{lv}(\tau)\}.$$

Thus $o \to o$ has level one, whereas $(o \to o) \to o$ has level two. A type level measures the ability to take functions as arguments; it does not measure the number of arguments. A curried type $o^k \to o$ has level one for every positive finite k.

The original calculus has variables, application and abstraction, together with zero, successor, case functionals and recursors. We use the types

$$0 : o, \quad S : o \to o, \quad D^\tau : o \to \tau \to \tau \to \tau, \quad R^\tau : o \to (o \to \tau \to \tau) \to \tau \to \tau.$$

In particular the order of the arguments to a recursor is fixed by

$$R^\tau 0AB \longrightarrow B, \qquad R^\tau (Sn)AB \longrightarrow A\, n\, (R^\tau nAB).$$

The case rules are $D^\tau 0AB \to A$ and $D^\tau (Sn)AB \to B$. Together with beta reduction these rules act inside arbitrary well-typed contexts. Terms in this paper are literal terms: an input is not first beta-normalized, uncurried, or replaced by an extensionally equal program.

A termination ordering is a relation that decreases on these reductions and is stable under contexts and substitutions. Recursive path ordering provides the corresponding first-order example. For ranked trees, $f(s_1, \ldots, s_k) > t$ follows either from a child dominating t, or from domination of all children of t together with a decrease of root precedence or a strict multiset decrease between equal roots. A strict multiset decrease replaces a nonempty collection by elements each smaller than a removed element. Extending this mechanism to typed application and abstraction introduces both type and binding information.

The separate ordinal-vector comparison studied below compares numerical assignments under *all* permitted substitutions. It is not the same definition as a recursive syntactic ordering. Nor is the height of a termination ordering the largest ordinal written in an assignment. The Bachmann–Howard calibration concerns the descending height of the syntactic relation. Its named ordinal notation systems and their comparison are supplied by the published ordinal analyses [5, 3, 4]; the exact new ordering and its proof are given in [8].

3 Ordinal expressions and the fixed numerical control

Every ordinal below ε_0 has a finite recursive Cantor normal form. Here ε_0 is the least fixed point of $\alpha \mapsto \omega^\alpha$. We write $\oplus, \otimes$ for natural sum and product: collect the Cantor monomials for addition, and multiply monomials by naturally adding their exponents for multiplication. These operations are commutative. Ordinary ordinal addition is retained when a Cantor normal form or a finite tail is displayed. For example $\omega \oplus 1 = \omega + 1$ but $1 + \omega = \omega$ for ordinary addition.

Definition 3.1 (Additive tree norm)**.** For a Cantor normal form with decreasing exponents and positive coefficients, put

$$\mathrm{N}(0) = 0, \qquad \mathrm{N}\left(\sum_{i<r} \omega^{\alpha_i} c_i\right) = \sum_{i<r} c_i(1 + \mathrm{N}(\alpha_i)).$$

Thus $\mathrm{N}(n) = n$, $\mathrm{N}(\omega) = 2$ and $\mathrm{N}(\omega^{\omega+1}) = 4$. Coefficients are charged with their full multiplicity, not their binary length.

Lemma 3.2 (Finite norm balls)**.** *Each norm ball is finite and can be effectively enumerated. Cantor-normal-form comparison, natural sum and natural product are effective. Moreover* $\mathrm{N}(\alpha \oplus \beta) = \mathrm{N}(\alpha) + \mathrm{N}(\beta)$.

Proof. An ordinal of norm at most b is represented by a finite rooted expression tree with at most b charged monomial nodes, including repeated monomials. There are finitely many such finite trees. Enumerate them, interpret them recursively, and retain their canonical Cantor forms. Alternatively enumerate by norm recursively: every exponent in a nonzero term of norm at most b has norm at most $b - 1$, so induction on b gives a finite list. Comparison is lexicographic comparison of exponent/coefficient lists, using the same recursive algorithm on exponents. The definitions of natural sum and product then give finite algorithms. Collecting repeated monomials does not change the sum of their charged norms, proving additivity. □

Fix throughout

$$F_0(t) = 2^t, \qquad F_{j+1}(t) = F_j^{t+1}(t), \qquad \Phi(t) = F_5(t+100), \tag{3.1}$$

where the exponent denotes iteration. Replacing the exponential base function by doubling would change every subsequent definition.

Lemma 3.3 (Elementary control properties)**.** *The fixed Φ is computable, strictly increasing, and greater than the identity. On its whole domain,*

$$\Phi(t+1) \geq 2^{2^{\Phi(t)}}.$$

In particular $\Phi(t+1) > 4\Phi(t) + 4$.

Proof. Induction on j proves that each F_j is computable, increasing, and strictly above the identity. For $n \geq 1$ the same induction gives $F_{j+1}(n) \geq F_j(n)$ and $F_1(n) \geq 2^{2^n}$. The $n+1$ evaluations inside $F_5(n+1)$ start from a strictly larger seed than those in $F_5(n)$. Since all values are integers,

$$F_5(n+1) = F_4(F_4^{n+1}(n+1)) \geq F_4(F_5(n)+1) \geq 2^{2^{F_5(n)+1}}.$$

Take $n = t + 100$. The stated weaker inequality follows. Its final consequence holds because $\Phi(t) \geq 4$ and $2^{2^a} > 4a + 4$ for $a \geq 4$. □

Definition 3.4 (Norm-controlled collapse)**.** For $\alpha < \varepsilon_0$ define, by ordinal recursion,

$$\psi(\alpha) = \max\big(\{0\} \cup \{\psi(\beta) + 1 : \beta < \alpha,\ \mathrm{N}(\beta) \leq \Phi(\mathrm{N}(\alpha))\}\big).$$

The set is finite by Lemma 3.2.

Lemma 3.5 (Controlled strictness and finite values)**.** *If $\beta < \alpha$ and $\mathrm{N}(\beta) \leq \Phi(\mathrm{N}(\alpha))$, then $\psi(\beta) + 1 \leq \psi(\alpha)$. Also $\psi(n) = n$ for finite n.*

Proof. The first assertion is one term of the defining maximum. For the second, induct on n. Every predecessor of the finite ordinal n is finite and has norm below $n \leq \Phi(n)$. The induction hypothesis makes the maximum equal to n. □

The further inequalities $\mathrm{N}(\alpha) \leq \psi(\alpha)$ and $k \leq \psi(\alpha + k)$, as well as the natural-product norm bounds used in source assignments, are Proposition 2.3 and Proposition 2.2 of Wilken–Weiermann [1]. Those are published background theorems, not assumptions about a new comparison algorithm. Controlled strictness does not assert that ψ is globally monotone in ordinal order.

4 A complete closed evaluator

Definition 4.1 (Maximum predecessor)**.** For $\alpha > 0$ and $b \in \mathbb{N}$ let

$$M(\alpha, b) = \max\{\beta < \alpha : \mathrm{N}(\beta) \leq b\}, \qquad L(e, b) = M(\omega^e, b).$$

The maximum exists because the set is finite and contains zero.

Theorem 4.2 (Effective maximum predecessor)**.** *The following simultaneous recursion computes M and L exactly. Write $\alpha = \omega^e c + r$ with $c > 0$ and $r < \omega^e$, and put $w = 1 + \mathrm{N}(e)$.*

1. *$M(\alpha, 0) = 0$ and $L(0, b) = L(e, 0) = 0$.*

2. *If $r > 0$ and $cw \le b$, retain the leading block: $M(\alpha, b) = \omega^e c + M(r, b - cw)$.*

3. *Otherwise set $d = \min(c - 1, \lfloor b/w \rfloor)$ when $r = 0$, and $d = \min(c, \lfloor b/w \rfloor)$ when $r > 0$. Then $M(\alpha, b) = \omega^e d + L(e, b - dw)$, omitting a zero block.*

4. *For $e, b > 0$, let $a = M(e, b - 1)$, $v = 1 + \mathrm{N}(a)$ and $(q, s) = \mathrm{divmod}(b, v)$. Then $L(e, b) = \omega^a q + L(a, s)$.*

Proof. If a nonzero tail can be decreased while retaining the entire leading block, every predecessor with a smaller leading block is inferior. The remaining budget and tail bound give clause 2. Otherwise the displayed d is the largest affordable coefficient still strictly below the original bound. After choosing that coefficient the tail must lie below ω^e, which is exactly the L query in clause 3.

For clause 4, any nonzero candidate below ω^e of norm at most b has leading exponent $a' < e$ with $\mathrm{N}(a') \le b - 1$. The largest possible exponent is therefore $a = M(e, b - 1)$. Lexicographic Cantor comparison first maximizes that exponent, then its coefficient q, then the remaining tail. The last tail is bounded by ω^a and has budget s, proving the formula.

For termination use the lexicographic pair (budget, phase), putting M above L at equal budget. Clause 2 lowers the budget since $cw > 0$. Clause 3 may keep it but lowers the phase. In clause 4, the M call uses $b - 1$, and $s < b$ because $v \le b$ and hence $q \ge 1$. Base cases terminate. This proves a finite algorithm and the asserted exactness, without an oracle for the maximum. □

Lemma 4.3 (Saturation at a limit)**.** *For every nonzero limit λ and $b \in \mathbb{N}$, $\mathrm{N}(M(\lambda, b)) = b$. The maximum strictly increases with b.*

Proof. If $\eta = M(\lambda, b)$ left unused budget $d > 0$, then $\eta + d < \lambda$ would have norm b and be larger. This contradicts maximality. For $b < c$, the ordinal $M(\lambda, b) + 1$ is below the limit and has norm $b + 1 \le c$. It witnesses strict increase. □

Theorem 4.4 (Single-predecessor recurrence)**.** *For every nonzero limit $\lambda < \varepsilon_0$ and $n \in \mathbb{N}$,*

$$\psi(\lambda + n) = \psi\big(M(\lambda, \Phi(\mathrm{N}(\lambda) + n))\big) + 1.$$

Proof. Induct on n. Put $b = \Phi(\mathrm{N}(\lambda) + n)$ and $\eta = M(\lambda, b)$. This is an admitted predecessor and has norm b, so its contribution gives the lower inequality. Any other admitted $\beta < \lambda$ satisfies $\beta \le \eta$. For $\beta < \eta$ the norm premise $\mathrm{N}(\beta) \le b = \mathrm{N}(\eta) \le \Phi(\mathrm{N}(\eta))$ and Lemma 3.5 bound its contribution by $\psi(\eta)$.

The remaining predecessors have the form $\lambda + j$, $j < n$. Their budgets $b_j = \Phi(\mathrm{N}(\lambda) + j)$ are strictly smaller than b. By saturation $\eta_j = M(\lambda, b_j) < \eta$, with $\mathrm{N}(\eta_j) = b_j < b$. Controlled strictness and the induction hypothesis give $\psi(\lambda + j) = \psi(\eta_j) + 1 \le \psi(\eta)$. Every term in the original maximum is therefore at most $\psi(\eta) + 1$. □

Corollary 4.5 (Closed evaluation)**.** *Every supplied closed collapse value is computable. At a nonfinite state $\alpha = \lambda + n$, replace α by $M(\lambda, \Phi(\mathrm{N}(\alpha)))$ and increment a counter. On reaching a finite state return that state plus the counter.*

Proof. Each replacement strictly decreases the ordinal. Well-foundedness below ε_0 proves termination. The recurrence proves preservation of the desired value, and Lemma 3.5 evaluates the last finite state. This is pointwise computability, not a uniform practical complexity bound and not a decision procedure for an infinite family of inputs. □

Theorem 4.6 (Complete terminal decomposition)**.** *Put $\nu = \mathrm{N}(\lambda)$ and let $t \geq \nu$ be the total initial norm. Starting at $\lambda + (t - \nu)$, let $\tau_\lambda(t)$ be the number of nonfinite replacements and $Z_\lambda(t)$ the first finite state. Then*

$$J_\lambda(t) := \psi(\lambda + t - \nu) = Z_\lambda(t) + \tau_\lambda(t), \qquad Z_\lambda(t) = \Phi^{\tau_\lambda(t)}(t).$$

The count is positive. In particular $Z_\lambda(t) < J_\lambda(t) < 2Z_\lambda(t)$.

Proof. Saturation makes every new norm exactly Φ of the old norm, including the last replacement into a finite state. Each replacement contributes one to the recurrence. At the finite endpoint the value equals its norm. There is no subsequent count of its finite predecessors. Since each application of Φ increases its argument by at least one, $Z_\lambda(t) \geq t + \tau_\lambda(t)$. A nonzero limit has $t \geq 2$; therefore $0 < \tau_\lambda(t) \leq Z_\lambda(t) - 2$ and the strict bounds follow. □

Corollary 4.7 (Finite coefficient identity)**.** *For finite k, n,*

$$\psi(\omega k + n) = \Phi^k(2k + n) + k.$$

Proof. The case $k = 0$ is finite evaluation. For $k > 0$, with $b = \Phi(2k + n) \geq 2k$, the maximum predecessor is $\omega(k - 1) + b - 2(k - 1)$. Apply the recurrence and induction on k. □

Lemma 4.8 (Strict separation in one finite tail)**.** *For $h_\lambda(n) = \psi(\lambda + n)$ with a fixed nonzero limit λ,*

$$h_\lambda(n + 1) > 2^{h_\lambda(n)}.$$

In particular every such function is injective.

Proof. Compare descents at total norms t and $t + 1$, using $t \geq \mathrm{N}(\lambda)$. Maximum predecessors are monotone in their ordinal bound and norm budget; dropping a finite tail also preserves ordinal order. Before the smaller descent reaches a finite state, induction on the simultaneous step number shows that the larger descent has not reached a finite state either. Its count L' is at least the smaller count $L \geq 1$. Writing $Z = \Phi^L(t)$, monotonicity and Lemma 3.3 give

$$\Phi^{L'}(t + 1) \geq \Phi^L(t + 1) \geq 2^{2^Z}.$$

For the last inequality, apply the control inequality to the last iteration; its two input integers differ by at least one. By the preceding theorem $h_\lambda(n) = Z + L < 2Z \leq 2^Z$. The larger value is at least its endpoint, which proves the strict inequality. □

The maximum-predecessor framework is classical [2]. The explicit recursion above fixes its exact specialization to this norm and collapse. The recurrence also holds for any computable strictly increasing control majorizing the identity. The very strong tail inequality uses the particular control (3.1).

5 The canonical source assignment

The assignment is a function of literal ordinal expression trees. We fix the following expansion convention throughout the collection. Every finite sum and positive integer repeat is a left-associated binary sum, preserving each operand as an unchanged subtree. A singleton sum is its operand. A numeral $k > 0$ is the corresponding repeat of one; an inner repeat remains nested when it is an operand of another sum. The sole different association is the explicitly parenthesized scalar box tail below. Exponential/product and collapse expressions

are primitive binary constructors with children f, g, rather than arbitrary product expressions to be reparsed.

An absent vector coordinate is a padding marker, interpreted as zero only for comparison. Vector addition omits an absent operand; it constructs a binary sum only when both operands are present. Likewise the optional finite-level contribution to the box is absent when its level is zero. This does not erase a literal zero from an input expression. Actual generated coordinates use the zero-free graded grammar below.

These choices make the printed finite-sum formulas explicit. They are not claimed to be the uniquely intended historical bracketing in [1]. More generally one may fix an ordered binary expansion at each macro site and arity, independently of the operands' names, values and internal trees, while preserving each operand. We call this an *admissible policy*. Uniqueness of the zero-history assignment is relative to that policy. Appendix A proves the domain and norm assertions for every such policy and exhibits a closed term whose assigned values differ between two policies. Unspecified reassociation is therefore not an innocuous preprocessing step.

For $A : \tau$ the canonical vector has indices $0, \ldots, \mathrm{lv}(\tau)$. Missing indices are padded by zero when vectors are compared. The prime clauses are

$$[\![0]\!] = (1), \quad [\![S]\!] = (1, 1), \quad [\![D^\tau]\!] = (1, \ldots, 1), \quad [\![R^\tau]\!] = (2, 1, \ldots, 1, \omega), \quad [\![X]\!] = \mathbf{x}.$$

For vector levels $m > n$, compute $h = f \square g$ downward by

$$h_i = f_i \ (i > n), \qquad h_i = 2^{h_{i+1}} \otimes (f_i \oplus g_i) \ (1 \le i \le n), \qquad h_0 = \psi(\omega \otimes h_1 \oplus f_0 \oplus g_0 \oplus n).$$

Application retains the coordinates through the result type's level. In the literal scalar tree we associate the tail as $f_0 + (g_0 + n)$ for $n > 0$ and as $f_0 + g_0$ for $n = 0$. This is relevant to syntactic partials.

For abstraction over $X : \sigma$, put $n = \mathrm{lv}(\sigma) + 1$ and let h have level m. Each auxiliary partial $\partial_i^X h_i$ is a vector of level n. The canonical zero-history rule is

$$[\![\lambda X.G]\!] = \delta^X [\![G]\!] + [\![G]\!]\{X := \mathbf{1}\},$$

where

$$(\delta^X h)_0 = 2^n \left(\sum_{i=0}^{m} \mathrm{sz}^X(h_i) \right) (\partial_0^X h_0)_0, \quad (\delta^X h)_j = \sum_{i=0}^{m} (\partial_i^X h_i)_j \ (0 < j \le n),$$

and $(\delta^X h)_j = h_j$ for $j > n$. The size is one at a whole X-free expression or variable leaf; an active sum or collapse adds the child sizes and one; $2^f \otimes g$ has size $2\,\mathrm{sz}^X(f) + \mathrm{sz}^X(g) + 1$.

At a whole X-free input u, $(\partial_i^X u)_j = u + 1$ if $i = j$ and is one otherwise. At x_i every output is one. Active sums add the partial vectors and $\mathbf{1}$. For $i > 0$,

$$\partial_i^X (2^f \otimes g) = 2\partial_{i+1}^X f + \partial_i^X g + \mathbf{1}.$$

At an active scalar $\psi(\omega \otimes f + g)$, the zeroth output is $\psi(\omega \otimes f\{X := \mathbf{1}\} + (\partial_0^X g)_0)$, and output $j > 0$ is $(\partial_1^X f)_j + (\partial_0^X g)_j$. The whole-free rule applies before any other rule, on the literal tree. These are the published clauses of [1, Definitions 2.10, 2.18 and 3.1]. The domain and bounded-norm assertions for the syntactic partial and abstraction rules are proved in Appendix A. The box domain and norm estimates use the value-level rules of [1] with the same primitive constructors. We do not assume a strict inequality $u\{X := \mathbf{1}\} < (\partial_0^X u)_0$: at $u = x_0$ both sides are one. The weak inequality suffices for domain closure, and the appendix proves the needed bounded-norm estimate directly with the printed multiplier.

The closed graded classes $\mathcal{B}_i$ are defined simultaneously as follows: one belongs to every class; ω belongs to classes of positive index; each class is closed under natural sum; $2^f \otimes g \in \mathcal{B}_i$ when $i \geq 1$, $f \in \mathcal{B}_{i+1}$ and $g \in \mathcal{B}_i$; $\psi(\omega \otimes f + g) \in \mathcal{B}_0$ when $f \in \mathcal{B}_1$, $g \in \mathcal{B}_0$ and $\mathrm{N}(f) \leq F_2(g)$; and a member of $\mathcal{B}_0$ may also be used in any positive class. These are classes of closed expression trees; zero is not a leaf of the grammar, although zero padding is used outside a vector's actual length. Admissible substitutions replace each entire variable vector by a closed vector with $x_i \in \mathcal{B}_i$ and $\mathrm{N}(x_i) \leq x_0$ for every positive index. A ground variable therefore ranges over all positive integers, independently of other names: every such integer is a finite sum of ones and there are no positive coordinates imposing a bound. Weak comparison is coordinatewise universal comparison. Strict comparison is strict at coordinate zero and weak at every positive coordinate; it is not lexicographic.

Theorem 5.1 (Actual-source forward translation)**.** *Fix an admissible expansion policy. For a literal original source with only ground free names, all positive canonical coordinates are closed and effectively evaluable. Coordinate zero is effectively a finite expression in positive-natural inputs, constants, addition and fixed functions $h_\lambda(n) = \psi(\lambda + n)$, where each direct index is $\lambda = \omega \otimes c > 0$ for a closed c. The translation preserves every shared input valuation, including arbitrary internal types, binders and recursors. It is effective uniformly in a supplied computable policy; it does not identify assignments made under different policies.*

Proof. Use simultaneous induction on source construction and the syntactic partial recursions. The invariant for a generated positive coordinate is that its parse uses positive-index variables, positive constants, natural addition and the graded exponential/product constructor, but no ground-index variable. The prime clauses have this property. The box computes positive fields using positive fields only and preserves it.

For a positive input to a partial, the whole-free rule retains only that positive input at its diagonal output and constants elsewhere. Thus it does not introduce a ground variable. At an active variable the result is constant. Sum and exponential clauses preserve the invariant recursively. For a scalar input, consider a positive output of its partial: the whole-free rule returns one, a tracked scalar variable returns one, and sums recurse. The collapse clause uses a positive-head partial and the positive output of the scalar-tail partial. These are exactly the two induction hypotheses. Hence even partials of scalar expressions cannot introduce ground-index variables into their positive outputs. The abstraction sums such outputs and substitutes ones for the bound vector in its retained part, so it preserves the invariant and removes all of that bound name's components.

Positive fields of the final vector can therefore mention only positive components of free term variables. There are none when every free variable is ground. Closed evaluation follows from the ordinal algorithms already proved. In particular the head of every final scalar collapse is closed.

At coordinate zero the prime and box clauses produce only constants, ground inputs, addition and collapse nodes. The scalar partial clause has the same form, and its size multiplier is a fixed syntactic integer, not a function of a valuation. Expanding it as a finite multiplicity preserves the grammar. Replace each closed-head collapse by $h_{\omega \otimes c}$. All replacements are compositional and use the original free names, proving the exact, valuation-preserving translation.

This argument concerns generated assignments. The ambient graded classes also allow promotion of certain whole-free scalar expressions to positive levels; the stronger invariant just proved is not asserted for all arbitrary members of those ambient classes. □

Corollary 5.2 (Two natural positive classes)**.** *For every fixed admissible policy, in a recursor-free source every positive coordinate is finite. If every internal type and subtype has level at*

most two, every direct scalar index is a polynomial limit below ω^{ω}.

Proof. Only the recursor prime introduces ω into a generated positive field. All other positive constructors, including the positive partial cases in the preceding proof, preserve finiteness after the final closures. This proves the first assertion for arbitrary internal type levels. For the second, coordinate two cannot contain an exponential constructor requiring coordinate three. Its generated closed values are linear Cantor polynomials $a\omega + b$. Coordinate one is built by sums and natural products using $2^{a\omega+b} = \omega^{a}2^{b}$, hence lies in $\mathbb{N}[\omega]$. Multiplication naturally by ω leaves a polynomial limit. A bound only on the output type would not bound internal coordinates in this way. □

The converse image assertion is not part of the translation: not every abstract expression need have an original source realizing it. Constructing such sources, retaining their finite box and input correlations, is one of the separate tasks addressed in [11].

5.1 A worked example

Let $I = \lambda z : o.z$ and $K = \lambda z : o.0$. The partial and size rules give $[\![I]\!] = (3, 1)$ and $[\![K]\!] = (5, 1)$: in the first case the scalar partial is one, in the second it is $1 + 1$, and the size multiplier is two. Thus, for a raw ground input $x_0 > 0$,

$$[\![Sx]\!]_0 = \Phi(x_0 + 3) + 1, \quad [\![Ix]\!]_0 = \Phi(x_0 + 5) + 1, \quad [\![Kx]\!]_0 = \Phi(x_0 + 7) + 1.$$

These are assigned values, not program denotations. Although $Ix \to x$, the canonical values differ. Canonical assignments fix the zero-history choice at every lambda; the normalization theorem in [1] can use history-dependent assignments. One cannot replace the input syntax by its normal form when deciding its canonical comparison.

6 How the companion results fit together

The collection divides the proofs according to their mathematical mechanism. The papers are supplied with their complete local arguments and build inputs; published background is cited conventionally. The following statements are an orientation to those proofs, not substitutes for them.

6.1 A syntactic termination ordering

The type-confined ordering paper [8] constructs a decidable, well-founded pair on original T together with a separate finite ranked ground signature. It is stable under well-typed contexts and substitutions, orients the original T rules, and restricts exactly to multiset recursive path ordering on the ranked ground subalgebra. Its individual descending ranks are below Bachmann–Howard and their supremum has that calibration.

The embedding of ranked nodes is part of the statement. A ranked ternary node and three curried applications of a native case functional are different trees. The theorem does not silently assert exact RPO on the latter encoding. Nor does it claim that arbitrary linear extensions of the partial ordering have its descending height, or that the relation is the canonical numerical comparison of Problem 1.1.

6.2 Identity and one shared input

In the residual grammar of Theorem 5.1, universal identity has a recursive normal form even with arbitrarily many independent inputs. The identity/one-variable paper [9] proves completeness by separating distinct open collapse tags on a suitable weighted ray. For one shared input it gives the stronger effective eventual-sign theorem: from $E(x), F(x)$ one obtains a threshold T and one sign that holds for all $x \geq T$. Closed evaluation handles the finite prefix.

Universal identity is not existential equality. Distinct functions may agree at particular inputs, and asking whether several such equalities have a simultaneous solution is not decided by the identity normal form. Likewise, two sources each having one free name do not form a one-input pair when their names are distinct. The new collapse-function results concern a different grammar from the positive exponential terms generated by $2^f g$; the earlier exponential comparison theorem does not by itself give the actual-source conclusions stated here.

6.3 Several inputs in recognized positive classes

The multivariate paper [10] proves the required arithmetic hypotheses for the fixed Φ and then treats finite coefficients, polynomial heads, and several larger structural classes. In particular, recursor-free actual sources are decidable without a bound on internal type level, and polynomial-head comparison covers actual sources whose maximum internal type level is at most two. These are consequences of the generated- source bounds in Corollary 5.2, not bounds merely on output type.

Common-head layers, balanced heterogeneous layers and single-core rows share some machinery, but retain different hypotheses. Their proofs keep every shared input correlation. A lower-level core occurring in two arguments cannot be replaced by two independent witnesses, and several arbitrary core values cannot be treated as independently variable just because their individual functions are computable.

6.4 Actual-source arithmetic and negative boundaries

The source-expressivity paper [11] defines an existential language whose additional operations are observations of *supplied actual sources*, rather than freely chosen ordinal labels. Its Boolean compiler turns an existential formula into one unconditional weak or strict canonical comparison, with the appropriate polarity. Absolute observers also handle sources of functional output type. Thus the remaining comparison problem and unsatisfiability in that actual-source language are computably equivalent.

The same paper separates two negative boundaries. At the unchanged control, functional free inputs already allow an undecidability reduction on actual sources. At all-ground inputs a different theorem obtains hardness after an additive-one perturbation of the control, even with agreement on an arbitrary prescribed finite prefix. The first changes the free-variable domain; the second changes the control. Neither settles Problem 1.1 by itself.

6.5 Exact counts and a moving partition coefficient

The counting paper [12] proves the retained-prefix transfer between actual norm-controlled descent counts and restricted partitions. For genuine linked parameters

$$u = \Phi(v), \quad B = \Phi(u), \quad C = \Phi(u+1), \quad d = B - u + 3, \quad e = u - 4,$$

one block's parity is

$$\delta_4(v) = 1 + p_d(C - e) \pmod 2, \qquad \sum_{n\ge 0} p_d(n) z^n = \prod_{i=1}^{d} (1 - z^i)^{-1}.$$

Both the cutoff d and the selected coefficient move. The complete Cartier observer retains every denominator factor, and its actual primitive-root residue tuple has an unbounded exact period. The theorem is about that tuple and the full state, not a lower bound for the selected scalar's complexity.

The proof of the growing-budget transfer matters: a constant-budget partition identity is not automatically an identity for a descent whose successive budgets follow Φ. Similarly a count stopped at a first ω^ω entry is not the complete count τ_λ of Theorem 4.6. The latter still includes every subsequent block.

7 The infinite-search obstruction

Each supplied ground tuple can be evaluated by a terminating computation. Consequently a false universal comparison has a semidecision procedure: enumerate tuples and stop at the first countervaluation. If there is none, this procedure has no reason to halt.

Proposition 7.1 (Countervaluation bounds)**.** *The fixed-control comparison problem is co-recursively-enumerable. It is decidable if and only if there is a total computable function $b(Q)$ such that each false comparison Q has a countervaluation whose every coordinate is at most $b(Q)$.*

Proof. Closed evaluation and enumeration prove the first claim. Given the stated bound, search the finite positive box and decide Q. Conversely, given a decider, return zero for a true query. For a false query enumerate tuples until its promised countervaluation is found, and return the largest coordinate. This search is entered only when its termination is known. With no free variables there is just one closed test. □

This elementary equivalence is useful precisely because it does not manufacture such a bound. A bound on evaluation time *at a given tuple* is not a bound on the location of the first countertuple. Larger ordinal upper bounds and faster evaluators can leave the latter problem unchanged.

The terminal decomposition gives another view of the obstruction. On a tail where Φ has stabilized residue r_m modulo a fixed m, proved with an effective threshold in [10],

$$J_\lambda(t) \equiv \tau_\lambda(t) + r_m \pmod m.$$

The enormous terminal value has not erased the small residue of the count. For example, eliminating an independent witness y from $\tau_\lambda(t) = 2y + 1$ asks whether an odd count occurs at all. It is not answered by the rate at which that count grows.

There is a genuinely necessary attainment problem for the complete count of a *supplied actual unary source*. Let $H : o \to o$ have vector (c_H, a_H), put $\lambda = \omega \otimes a_H$, and $C_H = \mathrm{N}(\lambda) + c_H + 1$. Its application satisfies

$$F_H(n) := [\![HX]\!]_0 - 1 = J_\lambda(C_H + n) - 1, \qquad x_0 = n + 1.$$

For fixed progression and modulus data, choose an effective starting point $b \ge C_H$ above the residue threshold, with $b \equiv a \pmod p$. On that progression the query

$$\exists t \ge b : \quad t \equiv a \pmod p, \quad \tau_\lambda(t) \equiv r \pmod m$$

becomes

$$\exists k, y \in \mathbb{N}: \quad F_H(b - C_H + pk) = my + s, \qquad s = (r + r_m - 1) \bmod m.$$

The finitely many earlier genuine arguments are evaluated separately. The actual-source compiler in [11] consumes the displayed formula. Thus a full comparison decider would decide this attainment query. The independent witnesses k, y explain why the one-shared-input theorem does not already decide it.

Certified periodic tables are a stronger proposed supplier for a particular general elimination route: one needs a threshold, period, exact residues and certification of all negative occurrence claims on the declared head family. The necessary supplied-source attainment problem should not be identified with tables for every auxiliary ordinal head. Nor has an extraction of the isolated δ_4 block from every complete source observation been proved. These distinctions prevent a diagnostic intermediate question from being mistaken for an equivalent reformulation of the whole problem.

8 Two small arithmetic lessons

8.1 An exact observer may already distinguish every numerator

Let $A(0) = 1$ and $\deg A = D$ over a field. On proper numerators the map

$$U \longmapsto ([z^0]U/A, \ldots, [z^{D-1}]U/A)$$

is invertible: its matrix is triangular with diagonal entries one. Hence distinct numerators have different coefficient series. In characteristic two their first differing coefficient has index below D and is selected by a binary continuation of length at most $\lceil \log_2 D \rceil$. This is the observability interpretation suggested by weighted-automaton theory [6].

It does not show that a particular source-admissible continuation distinguishes the states. An all-continuations minimizer and a compression preserving only the one actual remaining word are different problems. Nor may a rational Gram-matrix argument be imported unchanged over $\mathbb{F}_2$: the column $(1, 1)^T$ has rank one, while its Gram matrix is zero.

8.2 A successful cancellation in a neighboring problem

For $N = 2^k - 1$ and $1 \le t < N$, consider the finite set

$$\{(a, b) \in \{0, \ldots, N-1\}^2 : a + b \equiv t \pmod N,\ \mathrm{wt}(a) + \mathrm{wt}(b) < k\}.$$

Swap a, b. All off-diagonal objects cancel modulo two. There is one diagonal solution because two is invertible modulo N, and multiplication by two cyclically rotates the binary word. Its weight is $\mathrm{wt}(t)$. The set's parity is therefore $\mathbf{1}_{2\,\mathrm{wt}(t) < k}$. Only this elementary fact is used; a full proof of the Tu–Deng inequality is not assumed [7].

The restricted-partition problem has a different indexing set and different cancellation classes. The displayed involution therefore supplies no parity formula for it. The complete coefficient observer in [12] retains its entire generating function; a cancellation law for the selected coefficient would require an additional argument on that actual indexing set.

9 What would settle the remaining comparison problem

The established positive fragments exclude several tempting descriptions of the open region. It is not closed evaluation, universal identity, one shared ground input, arbitrary recursor-free syntax, or the internal-level-two fragment. The remaining region combines several independent ground inputs with nonpolynomial collapse observations beyond the recognized classes.

A positive completion needs a decision procedure for simultaneous existential actual-source observations, or another mechanism that certifies witness absence. A negative completion needs an exact interpretation of a known undecidable problem using the unchanged control, permitted source syntax and shared valuations. The source compiler already supplies the final conversion to one weak or strict comparison; another conjunction detector is not the missing theorem.

An unbounded full-state period does not refute a simple scalar law. An eventually periodic word is not automatically available effectively. A decidable word theory with unary labels does not automatically interpret addition on its positions. A source-construction theorem enlarges the represented arithmetic but does not decide that arithmetic's existential theory. These are distinct logical obligations, each requiring a theorem.

The purpose of the collection is to present the established mechanisms and their exact limits in ordinary mathematical form. The full fixed-standard- control, all-ground comparison problem remains open.

A Literal source syntax, policies and norm preservation

An ordinal expression in [1, Definition 2.1] is a literal tree. Its three binary constructors, with immediate subterms f, g, are

$$\mathrm{Add}(f,g), \qquad \mathrm{ExpProd}(f,g), \qquad \mathrm{Psi}(f,g).$$

Their interpretations are $f \oplus g$, $2^f \otimes g$ and $\psi(\omega \otimes f \oplus g)$, respectively. Here $\oplus, \otimes$ are natural sum and product. The last two expressions are primitive constructors, rather than arbitrary multiplication trees. In this note, ordinary arithmetic on natural-number sizes is distinguished from these literal ordinal trees.

For a tracked variable vector X, the size $\mathrm{s}_X(u)$ is one when the whole u is X-free, or when u is a component of X. Otherwise it is the sum of the child sizes plus one at Add and Psi, and twice the head size plus the tail size plus one at ExpProd. The partial operator also uses the whole-free case first. Thus

$$\mathrm{s}_X(x_0 + (1+1)) = 3, \qquad \mathrm{s}_X((x_0+1)+1) = 5.$$

The ordinal values of the two sums agree. This small example alone does not show that the distinction occurs in a generated source assignment; Section A.4 supplies such an assignment.

The source specifies finite sums and natural-number multiples in Definitions 2.10 and 2.18 without a full binary association convention. Its convention for left-associated lambda-term application concerns a different grammar. Also, actual graded coordinate trees are zero-free, whereas out-of-range vector components are written as zero. Footnote 5 to Definition 2.10 expressly explains that the box definition is designed to avoid zero occurrences in coordinate parse trees. A literal implementation must respect both facts.

A.1 Explicit finite parsing policies

Definition A.1. An admissible policy ρ is a computable rule choosing an ordered full binary addition tree for each nonempty finite-sum macro site and its arity. Each operand is used once as an unchanged subtree. The choice is independent of operand names, values, internal syntax and free-variable status. A k-fold repeat, $k > 0$, has exactly k unchanged copies of its operand. Existing subtrees are never flattened, normalized or reassociated.

The macro sites are finite numerals; positive-integer repeats; the three-operand active-sum partial; the three operands $\mathrm{Repeat}_2(\partial_{i+1}f), \partial_i g, \mathbf{1}$ in an active exponential partial; the positive-coordinate abstraction sums; its scalar repeat; and the optional three-operand box tail. Repeat2 remains a single nested operand in the exponential clause. Binary clauses stay binary. The integer

$$S_X(h) = 2^n \sum_{i=0}^{m} \mathrm{s}_X(h_i), \qquad n = \mathrm{lv}(X) + 1,$$

is computed in natural-number arithmetic from the already fixed input trees, then used as a repeat count.

Each vector records its actual level. An out-of-range lookup has an *absent* role, interpreted as zero for comparison. A vector sum omits absent operands but retains every actual operand unchanged. The box's optional finite-level contribution is absent when its argument level is zero; it is a positive numeral otherwise. This is not a rule that simplifies actual Zero nodes or tests whether an expression has value zero. Every actual source coordinate remains zero-free. In particular the source program 0 is assigned One.

A useful default takes left combs for sum, repeat, numeral and positive aggregation macros, including the active-sum partial, while retaining the already explicit box tail $f_0 + (g_0 + n)$ for $n > 0$ and $f_0 + g_0$ for $n = 0$. This is a declared convention, not a claim that the source uniquely selected it.

Proposition A.2. *For a fixed policy, all literal partials and zero-history assignments are uniquely determined finite trees. Their expansions have computable bounds in the input trees and type levels. For a fixed input tree, changing only the output associations of a partial does not change its ordinal value.*

Proof. Structural recursion determines a partial. Each zero-history lambda then uses its body's fixed tree, its fixed sizes, and the finite repeats specified by the policy. Erasing only the newly supplied parentheses recovers the printed formulas exactly.

For an explicit bound, let $|u|$ count tree nodes. Give a leaf weight one, Add/Psi weight $w(f) + w(g) + 1$, and ExpProd weight $2w(f) + w(g) + 1$. Induction gives

$$\mathrm{s}_X(u) \le w(u) \le 2^{|u|} - 1.$$

Every coordinate of a partial has at most $E(|u|) = 6 \cdot 3^{|u|}$ nodes. At a whole-free diagonal it has $|u| + 2$ nodes. At active Add, ExpProd, scalar Psi and positive Psi outputs the respective bounds are

$$E(f) + E(g) + 3, \quad 2E(f) + E(g) + 4, \quad |f| + E(g) + 1, \quad E(f) + E(g) + 1,$$

where $E(f)$ abbreviates $E(|f|)$. These fit $E(1 + |f| + |g|)$. Thus $S_X(h) \le 2^n \sum_i (2^{|h_i|} - 1)$. The scalar repeat has $S_X(h)(|(\partial_0 h_0)_0| + 1) - 1$ nodes; a positive abstraction sum has at most $\sum_i E(|h_i|) + m$ nodes. The box is a finite downward recursion, with $|b_i| = |b_{i+1}| + |f_i| + |g_i| + 2$ at $1 \le i \le n$. These bounds do not evaluate ψ.

For fixed input trees, structural induction on the partial clauses uses associativity of natural sum and the same finite multiplicities to prove output-value independence. An enclosing partial need not see identical input trees, so this does not prove source-policy independence. □

A.2 The weak scalar lemma and the graded domain

The domain C_i^X is as in [1, Definition 2.9]. Actual coordinate trees are positive. Its scalar class has finite values; the natural norm of a finite value is that value. Write $P_i = \partial_i^X$ and $n = \mathrm{lv}(X) + 1$. All expression inequalities below hold under every admissible substitution.

Remark A.3. Printed Lemma 2.19, page 22 of [1], asserts $h\{X := \mathbf{1}\} < (P_0 h)_0$ for all $h \in C_0^X$. At $h = x_0$ both sides equal one. This is a false strict assertion in the actual PDF, not an extraction artifact. The appendix's strict estimate (A.5) also fails: for $\mathrm{lv}(X) = 0$, $h = (x_0, 1)$, $i = 1$ and the free subterm $t = 1$, it asserts $1 < 1$.

Theorem A.4 (Weak scalar replacement)**.** *For $h \in C_0^X$,*

$$h\{X := \mathbf{1}\} \le (P_0 h)_0.$$

Every partial $P_i h$, $h \in C_i^X$, is a valid X-free C-vector of level n. For a vector $h \in C^X$, the assembled $\delta^X h$ is likewise valid and X-free. These assertions hold for every admissible policy.

Proof. Induct simultaneously on the grammar and the partial recursion. A whole-free scalar has partial $h + 1$, a tracked scalar variable has partial one, and an active sum has scalar partial $P(f) + P(g) + 1$. At an active collapse the two heads after $X := \mathbf{1}$ are identical, and the induction hypothesis compares the finite tails. For a fixed ordinal a and finite r, the published inequality $\psi(a) + r \le \psi(a + r)$, Proposition 2.3(1), gives the needed weak finite-tail monotonicity. No global monotonicity of ψ is used.

For the new collapse's domain premise, the original premise and the weak induction hypothesis give

$$\mathrm{N}(f\{X := \mathbf{1}\}) \le F_2(g\{X := \mathbf{1}\}) \le F_2((P_0 g)_0).$$

Its head has the correct positive grade. Whole-free diagonal outputs are $h + 1$, off-diagonal outputs and tracked-variable outputs are One. The other cases only sum correctly graded partial outputs, with the stated finite multiplicities. Positive collapse outputs add the head partial and the positive tail partial. Each output is X-free. Assembling the finite sums and the positive scalar repeat preserves the domain. A coordinate above n is already X-free: there is no tracked component at its grade, and positive promotion of a scalar requires whole-X-freeness. It may be retained unchanged. □

A.3 Bounded norm with the original multiplier

The additive norm satisfies the published inequalities

$$\mathrm{N}(a \oplus b) = \mathrm{N}(a) + \mathrm{N}(b), \quad \mathrm{N}(a) + \mathrm{N}(b) - 1 \le \mathrm{N}(a \otimes b) \quad (a, b > 0), \qquad \mathrm{N}(a) \le 2\,\mathrm{N}(2^a).$$

Also $\mathrm{N}(a) \le \psi(a)$ and $r \le \psi(a + r)$ for finite r. These are Propositions 2.2 and 2.3 of [1].

Lemma A.5. *Let h be bounded in norm. Fix an admissible substitution for its other names and put $B = h_0\{X := \mathbf{1}\}$. Every first wholly X-free subtree encountered at a positive grade in a partial recursion starting at any actual coordinate of h satisfies*

$$\mathrm{N}(t) \le 2^{n-1} B.$$

Proof. Compute norms after $X := \mathbf{1}$; the free t is unchanged and every input coordinate then has norm at most B. Descending through an Add or an exponential tail does not increase norm. An exponential-head step costs at most two, since

$$\mathrm{N}(f) \le 2\,\mathrm{N}(2^f) \le 2\,\mathrm{N}(2^f \otimes g),$$

and increases grade by one. At grade n a subtree is already wholly X-free and recursion stops. From a positive root there are at most $n-1$ such steps.

A scalar path can enter a positive grade only through a collapse head. This costs no factor two: positivity and the cited bounds give

$$\mathrm{N}(f) \le \mathrm{N}(\omega \otimes f \oplus g) \le \psi(\omega \otimes f \oplus g), \qquad g \le \psi(\omega \otimes f \oplus g).$$

The head enters grade one, leaving at most $n-1$ active exponential-head steps. Scalar tails stay at grade zero. A promoted scalar at a positive grade is already whole-free, so cannot cause an additional active descent. This proves the bound. □

Theorem A.6. *On vectors $h \in C^X$, the operator δ^X preserves bounded norm with its original multiplier $S_X(h) = 2^n \sum_i \mathrm{s}_X(h_i)$, for every admissible policy.*

Proof. Continue at an arbitrary fixed substitution for the other names. Set $H = (P_0 h_0)_0$, so $1 \le B \le H$ by Theorem A.4, and put $\nu = 2^n H$. A free positive-grade subtree which can be copied into a positive diagonal output satisfies

$$\mathrm{N}(t) + 1 \le 2^{n-1} B + 1 \le 2^n H = \nu.$$

A free scalar subtree is not copied into a positive output; those outputs are One. This distinction pays the extra diagonal unit without the source's false strict estimate (A.5).

Induct inside each partial to obtain, for $0 < j \le n$,

$$\mathrm{N}((P_i h_i)_j) \le \mathrm{s}_X(h_i)\nu.$$

The whole-free and tracked-variable cases follow from the preceding paragraph. For Add the sum of the two bounds plus one is at most $(\mathrm{s}_X(f) + \mathrm{s}_X(g) + 1)\nu$. For ExpProd there are two copies of the head bound, one tail bound and One, giving $(2\,\mathrm{s}_X(f) + \mathrm{s}_X(g) + 1)\nu$. A positive collapse output only adds the head and tail bounds. Norm additivity makes all associations valid. Summing over i gives

$$\mathrm{N}((\delta^X h)_j) \le \Big(\sum_i \mathrm{s}_X(h_i)\Big) 2^n H = (\delta^X h)_0.$$

For $j > n$ the unchanged h_j is X-free, so its norm is at most $B \le H \le (\delta^X h)_0$. This holds at every substitution. □

A.4 An actual closed source separates the policies

Let L and R agree at every macro except the active-sum partial, where they choose respectively $(a + b) + 1$ and $a + (b + 1)$. Repeats in both are left combs. Consider the literal source

$$A = \lambda z : o.\lambda x : o.((\lambda w : o.z)\, x) : o \to o \to o.$$

It is closed and recursor-free; every internal type has level at most one. There is no beta normalization before assignment.

The assignment of $\lambda w.z$ is $(f(z), 1)$, with the literal tree

$$f(z) = ((z+1)+(z+1))+z, \qquad \mathrm{s}_z(f) = 9.$$

The x-body scalar is $g = \mathrm{Psi}(1, f(z)+x)$, whose x-size is five and whose abstraction multiplier is ten. Put

$$u = f+1, \quad T_L = (u+1)+1, \quad T_R = u+(1+1), \quad K_\rho = \mathrm{Psi}(1, T_\rho), \quad B_* = \mathrm{Psi}(1, u).$$

The x-abstraction has

$$h_0 = \mathrm{Repeat}_{10}(K_\rho) + B_*, \qquad h_1 = (1+1) + \mathrm{Sum3}_\rho(1,1,1).$$

The latter is wholly z-free and has value five. Literal sizes and positive partial values are as follows:

	f	u	T_ρ	K_ρ	$\mathrm{Repeat}_{10}(K_\rho)$	B_*	h_0
s_z under L	9	11	15	17	179	13	193
s_z under R	9	11	13	15	159	13	173

At every scalar expression in this table, output one of its z-partial has the corresponding size as its finite value. Whole-free scalar inputs give one, a tracked variable gives one, active sums add the two values and one, and each collapse adds the head partial's value two to its tail's value. Thus the positive output of $P_0 h_0$ is 193 or 173. Since $(P_1 h_1)_1$ has value six, the outer abstraction gives

$$\boxed{[\![A]\!]_{1,L} = 193+6+5 = 204, \qquad [\![A]\!]_{1,R} = 173+6+5 = 184.}$$

These are distinct assigned ordinal values, not merely distinct trees.

The scalar values also separate. Write $H(t) = \psi(\omega + t)$. Direct scalar partial calculation gives $P_0 f = 11$, $P_0 u = 14$, and $P_0 T_L = 20$, $P_0 T_R = 18$. The outer size multipliers are 388 and 348; the common retained value is $10H(8) + H(6)$. Consequently

$$[\![A]\!]_{0,L} = 388(10H(20) + H(14) + 10) + 10H(8) + H(6),$$
$$[\![A]\!]_{0,R} = 348(10H(18) + H(14) + 10) + 10H(8) + H(6).$$

The source's finite-tail inequality gives $H(20) > H(18)$, so the L scalar is strictly larger too. No numerical evaluation of the control or collapse is needed. The accompanying bounded AST control reconstructs the typed source and confirms the finite 204/184 coordinates exactly.

A.5 What the forward translation preserves

Theorem A.7. *Fix an admissible policy. For a literal original source all of whose free term names are ground, every positive final coordinate is closed and effectively evaluable. Its scalar coordinate translates effectively, preserving every shared valuation, to an expression in positive-natural inputs, positive constants, addition, and fixed functions*

$$h_\lambda(t) = \psi(\lambda + t), \qquad \lambda = \omega \otimes c > 0 \quad (c \textit{ closed}).$$

Arbitrary internal binders, types and recursors are allowed.

Proof. Induct simultaneously on source assignments and partial recursions. A generated positive field contains only positive-index components of variables, positive constants, Add and the graded ExpProd constructor. In particular it contains no component of index zero of any name. Primes and positive box fields preserve this stronger invariant. A positive whole-free

partial copies only its positive input at the diagonal and returns constants elsewhere. Active-variable outputs are constant; Add and ExpProd partials preserve the invariant. A scalar partial's positive outputs are constants in the whole-free and variable cases, recurse at sums, and at a collapse add a positive-head partial and a positive output of its scalar-tail partial. These are exactly the simultaneous induction hypotheses. Abstraction sums these outputs and removes all components of its bound name, including in its retained part.

The only positive-index variables that could remain are positive components of free source names. There are none when those names are ground. All final positive fields, including scalar-collapse heads, are therefore closed. Their positivity follows from the graded domain proved above. Scalar generated fields contain only ground components, constants, Add and Psi. Replace each closed-head Psi by the displayed h_λ applied to its recursively translated tail. The syntactic size multipliers are fixed integers, expanded as finite additions; they do not introduce fresh independent inputs. This proves the exact shared-valuation statement. □

Only the recursor prime introduces ω into generated positive fields. Thus in a recursor-free source the final positive fields are finite. If all internal types and subtypes have level at most two, coordinate two is built by sums from ω and finite constants, so is $a\omega + b$. Coordinate one then lies in $\mathbb{N}[\omega]$, since $2^{a\omega+b} = \omega^a 2^b$. Every direct scalar index is consequently below ω^ω. These assertions hold for each policy; the example above shows that they do not imply policy independence.

A.6 Limits of the repair and later uses of strictness

Theorem A.4 pays the domain use immediately following printed Lemma 2.19. Theorem A.6 replaces the false strict (A.5) in Appendix A.2 and proves its required (A.6) and bounded-norm conclusion with the same constant. Its higher-coordinate use needs only $\mathrm{N}(h_j) \le B \le H \le (\delta h)_0$.

Appendix A.7(2) also invokes the false strict link $F_2(\bar{g}) < F_2((P_0 g)_0)$. In its active-collapse induction, let $Q = \psi(\omega\, \mathrm{s}_X(g)\alpha + \mathrm{s}_X(g)m)$ be the strict upper bound on $(P_0 g)_0$ from the induction hypothesis. The repaired chain is

$$\mathrm{N}(\bar{f}) \le F_2(\bar{g}) \le F_2((P_0 g)_0) < F_2(Q).$$

The needed strict slack is supplied by that upper-bound induction hypothesis, not by the false lemma. This describes that local repair; it is not a fresh audit of every quantitative appendix estimate.

The distinct beta supplier, Lemma 2.22, compares $h < (P_i h \Box X)_i$. Its variable case is strict because of the box and the extra input. It does not assert the false strict scalar lemma without the box. The original termination proof uses history-dependent assignment derivations. It does not assert that fixed zero-history values decrease under all reductions.

The results here establish exact, policy-relative domains, norms and the forward translation. They do not supply a full comparison decision procedure, reconstruct a unique historical policy, or assert that all extended annotated-recursion histories have the stronger generated invariant of Theorem A.7.

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